INCH-POUND

MIL-P-70786 (AR)
29 June 1989

MILITARY SPECIFICATION

PALLET, STANDARD METAL

This Specification is approved for use, within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture of standard metal pallets for use in unitizing ammunition (see 6.1).

1.2 Classification. The standard metal pallets shall be of the following types (see 6.2):

Type I - Pallet, Standard Size 44" x 40", Sheet Metal
Type II - Pallet, Standard Size 45 1/2" x 35", Sheet Metal
Type III - Pallet, Standard Size 48" x 40", Sheet Metal
Type IV - Pallet, Standard Size 53" x 42", Sheet Metal

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning and Preparation of Ferrous and Coated Surfaces for Organic Protective Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8140
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MIL-W-12332	- Welding Resistance, Spot, Seam and Production, for Fabricating Assemblies of Low Carbon Steel
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-70625	- Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval of

STANDARDS

FEDERAL

FED-STD-101	- Test Procedures for Packaging Materials

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-1235	- Single and Multi Level Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1261	- Welding Procedures for Constructional Steel
MIL-STD-1660	- Design Criteria for Ammunition Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, Pa 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in affect on the date of the solicitation.

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DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

ACL00000423	-	CARC Finishing Requirements for Metal- and Pallet Adapters
AC200000431	-	Pallet, Standard Size 44" x 40", Sheet Metal
AC200000433	-	Pallet, Standard Size 45 1/2" x 35", Sheet Metal
AC200000435	-	Pallet, Standard Size 48" x 40", Sheet Metal
AC200000437	-	Pallet, Standard Size 53" x 42", Sheet Metal

(Copies of specifications, standards, drawings and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

ANSI/AWS B 4.0 - Mechanical Testing of Welds, Standard Methods for

(Application for copies of AWS publications should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, Florida 33135.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Pallets. The pallet assemblies shall comply with all requirements specified on drawings AC200000431, AC200000433, AC 200000435 and AC200000437, as applicable and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Stacking strength. The pallets shall withstand a 16,000 pound minimum (min.) stacking load.

3.4 Welding. The welding shall comply with the requirements of MIL-W-12332 or MIL-STD-1261 and weld callouts in the applicable drawings.

3.5 Weld integrity.

3.5.1 Visual inspection. The load bearing surface welds (see applicable dwg.) of the pallet assembly shall be free of the following defects:

a. Undercut.

b. Overlap.

c. Surface cavities.

d. Surface cracks in weld metal or in heat affected zone of base metal.

e. Lack of weld penetration.

In addition, fine scattered (average pore diameter equal to or less than 0.030 inches) porosity exceeding 13 pores in any one square inch shall be rejected. Not more than two pores may exceed 0.030 inches in diameter. Coarse scattered (average diameter greater than 0.030 inches) porosity exceeding two pores in any one

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square inch of weld face shall be rejected. No single pore may exceed 0.060 inches in diameter. Linear surface porosity in excess of 0.500 inches or in excess of 25 percent of total length, whichever is less shall be rejected.

3.5.2 Weld coupon bend test. The guided bend specimens shall have no open discontinuities exceeding 3.2mm (1/8 in.) measured in any direction on the convex surface of the specimen after bending, except that cracks occurring on the corners shall not be considered, unless there is definite evidence that they result from slag inclusions or other internal discontinuities.

3.6 Painted surface. The painted surface shall comply with 3.6.1 or 3.6.2, as applicable.

3.6.1 Conventional coating. The painted surfaces of the pallets shall comply with the requirements of the applicable drawings, TT-C-490 and with instructions in MIL-STD-171.

3.6.2 Chemical agent resistant coating (CARC). The painted surfaces of the pallets shall comply with the requirements of dwg. AC200000423 and applicable drawings.

3.7 Unit load tests. The unit loaded pallets shall conform to the requirements specified in MIL-STD-1660 (see 4.5.3).

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship.

3.9.1 Pallets. The pallets shall be regular, smooth, free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability and safety of the pallets. Weld splatter is permissible on non-load bearing surfaces provided it is firmly attached.

3.9.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided there are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total coverage. Paint chips which expose primer or bare metal are acceptable provided they are not more than 0.25 inches in diameter and no more than two such paint chips on a pallet. Scratches which expose primer or bare metal are permissible provided they are no more than 1.5 inches in length and no more than 10 inches total on a pallet. Bare metal on contact surfaces of bottom of skids and in areas inaccessible to paint is permissible.

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3.9.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance, of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of twelve (12) painted pallet assemblies, ten (10) unpainted pallet assemblies and six (6) each of every subassembly and component.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70786 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Assemblies and components				As applicable
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Stiffener (Dwg. AC200000427)				
	a) Overall width	6		3.2	Gage
	b) Overall length	6		3.2	Gage
	c) Evidence of poor workmanship	6		3.9	Visual
	d) Material thickness	3		3.2	Gage
	Skid (Dwg. AC200000428)				
	a) Overall width	6		3.2	Gage
	b) Overall length	6		3.2	Gage
	c) Evidence of poor workmanship	6		3.9	Visual
	d) Material thickness	3		3.2	Gage
	Post (Dwg. AC200000429)				
	a) Height	6		3.2	Gage
	b) Evidence of poor workmanship	6		3.9	Visual
	c) Material thickness	3		3.2	Gage
	Deck (Applicable drawing)				
	a) Overall width	6		3.2	Gage
	b) Overall length	6		3.2	Gage
	c) Evidence of poor workmanship	6		3.9	Visual
	d) Material thickness	3		3.2	Gage

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70786 (AR)

PARAGRAPH	TITLE: Assemblies and components		SHEET 2 OF 2		DRAWING NUMBER: As applicable
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Unpainted pallet assemblies (Applicable drawing)				
	Examination for defects	10		3.2	4.4.2.1
	Pallet assemblies (applicable dwg)				
	a) Examination for defects	6		3.2	4.4.2.2
	b) Stacking strength	9		3.3	4.5.1
	c) Unit load test	3		3.7	4.5.3

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces AMSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation of MIL-A-48078.

4.4.2 Examination. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70786 (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.1	Unpainted pallet assembly		1 1		AC200000431, 433, 435 or 437, as applicable NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mating surface of weld incorrect (Note 1)		1.0	3.2	Gage
102	Location of adapter mounting hole(s) in deck		1.0	3.2	Gage
103	Location of adapter mounting hole(s) in skid		1.0	3.2	Gage
104	Height of pallet		1.0	3.2	Gage
105	Overall length		1.0	3.2	Gage
106	Overall width		1.0	3.2	Gage
107	Weld without wrap (Note 1)		1.0	3.2	Visual
108	Visual inspection of weld, unsatisfactory (Note 1)		1.0	3.5.1	Visual
Minor					
201	Evidence of poor workmanship		1.5	3.9.1, 3.9.3	Visual

NOTES:

pertain to load bearing welds. These are the welds which join the post to the skid and the post to the deck.

AMSMC Form 1570, 1 Feb 85 Replaces AMSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70786 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER AC200000431, 433, 435 or 437, as applicable
4.4.2.2	Pallet assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protective coating thickness (total thickness minus primer thickness (see dwg. AC200000423 and Notes 1 and 2 below)).		1.0	3.2	Gage
Minor					
201	Protective finish missing, incomplete or incorrectly applied		1.5	3.9.2	Visual
202	Marking incorrect or missing		1.5	3.2	Visual
203	Evidence of poor workmanship (other than protective finish)		1.5	3.9	Visual

NOTES:
1. This inspection only applies when the assembly has been painted with CARC.
2. The primer thickness shall be determined from a test panel.

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4.4.3 Testing.

4.4.3.1 Stacking strength. (See 3.3), Major defect - The pallets shall be selected in accordance with the appropriate sampling plan as follows:

Lot Size	Sample Size
1-200 pallets	2
201-500 pallets	5
501-1500 pallets	8
1501-or more pallets	12

The samples shall be tested in accordance with 4.5.1. If any sample fails to meet the requirement, the lot shall be rejected. After three (3) consecutive lots have been found acceptable, every third lot shall be tested thereafter. If any of these lots are rejected, testing must be returned to three (3) consecutive lots. After three consecutive lots pass, return to testing every third lot.

4.4.3.2 Weld coupon bend test. (See 3.5.2), Major defect - During the first half of the production shift and during the last half of the production shift and whenever the electrodes are changed or dressed and when any changes are made in the control of the machine, two weld coupons shall be prepared by the welder in accordance with ANSI/AWS B4.0. If any test coupon fails to meet the requirement, when tested in accordance with 4.5.2, the quantity of pallets welded during that shift and after the earlier weld coupon test (if applicable) shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in Table I., 4.4.2.1, 4.4.2.2 and in 4.5. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. When Automated Acceptance Inspection Equipment is to be used, the provisions of MIL-A-70625 shall apply. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Stacking strength. The pallet shall be stacked with the required stacking load for a period of one (1) hour minimum. After that time, the pallet shall not have any evidence of breaking, bending, buckling or failure of any of the welds. Any pallet which fails to meet the requirement shall be classed defective.

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4.5.2 Weld coupon bend test. A guided bend test, as specified in ANSI/AWS B4.0, shall be performed on the coupons. Any coupon which fails to meet the requirement, shall be classed defective.

4.5.3 Unit load test. (First article inspection only) Prototype unit loads for the pallets shall be fabricated in accordance with MIL-STD-1660 with the following exceptions.

The gross weight of the ammunition load shall be 4,000 pounds minus 50 pounds, evenly distributed over the deck and shall obtain a height of 48 inches ± 2 inches. The following tests, on the loaded pallets, shall be performed in accordance with MIL-STD-1660.

4.5.3.1 Repetitive shock test. The acceptance/rejection criteria specified in section 7 of Method 5019.1 in FED-STD-101 is mandatory.

4.5.3.2 Edgewise drop (rotational) test. The acceptance/rejection criteria specified in section 7 of Method 5008.1 in FED-STD-101 is mandatory.

4.5.3.3 Incline impact test. The acceptance/rejection criteria specified in section 7 of Method 5023 in FED-STD-101 is mandatory.

4.5.3.4 Sling compatibility test. The acceptance/rejection criteria specified in section 5 of MIL-STD-1660 shall apply.

5. PACKAGING

5.1 Preservation and packing. None required.

5.2 Packing.

5.2.1 Commercial. The metal pallet shall be packed in accordance with ASTM-D3951.

5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM D3951-82 and the following additional instructions. No shipping marking shall be placed directly on the pallets. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If pallets are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

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5.4 Shipping. When pallets from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

6. NOTES

6.1 Intended use. This metal pallet is intended for use with ammunition containers, projectiles and cartridges.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision, and drawings of any special equipment required.

6.5 Subject term (key word) listing.
Ammunition
Cartridge
Container
Metal
Pallet
Projectiles

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 8140-A767)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70786 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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